Theme: Explainability in Machine Learning

- ML has become an important tool for studying predictability.eg:return
- Compared with traditional linear models, ML models can capture:
  - high-dimensional predictors;
  - nonlinear pricing relations;
  - interactions among bond, firm, and macro variables.
- However, stronger predictability often comes with a major concern: the black-box nature of ML.
- Therefore, the key issue is not only whether ML can predict returns, but also why it predicts.

Theme: Explainability in ML-现有可解释性方法

1. 高可解释性模型

线性回归 | 线性ML如logic

- **最直观：系数大小 – 影响程度；系数方向 – 影响方向**
- 假设：特征独立；线性关系

2. 特征重要性

树模型 | 非线性ML

- **量化每个特征对预测精度的贡献程度**
- 难以识别"如何影响"； 事后解释

机器学习可解释性：Explainable

3. SHAP 非线性ML

- **基于博弈论；精度贡献可识别方向性**
  全局+局部，衡量特征重要性
- 计算慢；事后解释（对黑箱行为的模拟）

4. LIME 非线性ML

- **局部点建立线性模型，提供解释力**
- 非全局关系；事后解释

Glass Box--EBM: 事前建立内在可解释性模型：

$$r_{t+1} = \beta + f_1(x_1) + f_2(x_2) + \cdots + f_{12}(x_1, x_2) + \epsilon$$

Glass Box Machine Learning and Corporate Bond Returns

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(Journal of Financial Economics, 2026)

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2026 年 6 月 27 日

Question

- Can glass-box machine learning predict corporate bond returns as well as black-box models?
- Can it reveal the economic structure behind return predictability?
  - Which bond, firm, and macro variables matter?
  - Are the pricing effects nonlinear?
  - Which interactions are economically important?
- Can interpretability improve portfolio construction?

Why interesting?

- **Factor zoo:** Corporate bond returns are related to many risk factors.
- **ML advantage:** handle high-dimension, capture nonlinear and interactions.
- **Black-box problem:** More complex models improve prediction, but make it difficult to understand Why Priced.
  - Ensemble models make it hard to see each variable's contribution.
  - Complex models capture nonlinear, but economic mechanism remains opaque.
- **This paper:** EBM, a glass-box model: explainable structure + ML flexibility
  - reveals nonlinearities and interactions that explain corporate bond returns.

Contribution

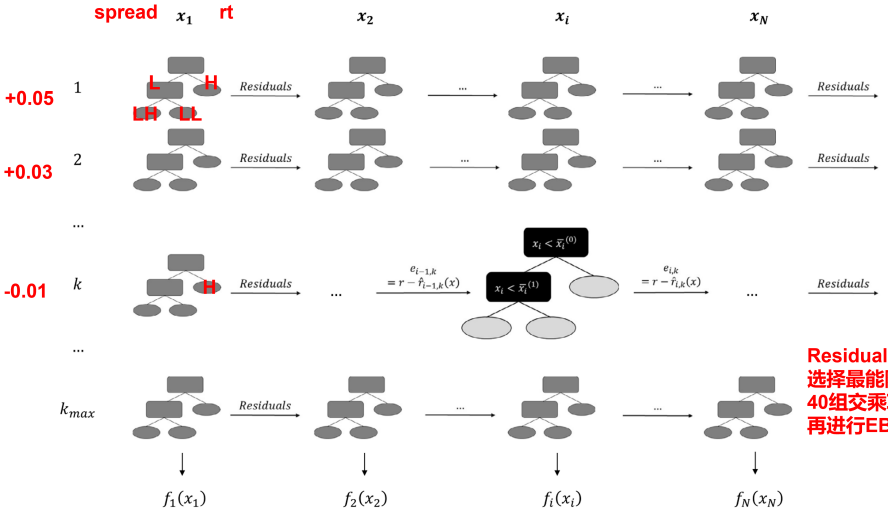
- **Novel methods in asset pricing:**
  - Prior ML studies emphasize forecast accuracy using NN, tree, PCA $\cdots$(Gu et al., 2020; Kelly et al., 2023).
  - **This paper:** EBM:combine explainability with strong predictive performance.
- **Investigating bond return predictors:**
  - Prior bond studies document predictors such as size, credit risk, value, momentum, liquidity, reversal, and macro uncertainty(Bali et al., 2021a,b)
  - **This paper:** reveals nonlinear interactions between macro and bond factors.

Design–Explainable Boosting Machine

- EBM is based on a generalized additive model with selected pairwise interactions:

$$r_{l,t+1} = \beta + \sum_{i=1}^{N} f_i(x_{i,l,t}) + \sum_{i>j} f_{ij}(x_{i,l,t}, x_{j,l,t}) + e_{l,t+1}.$$

- $f_i(\cdot)$ captures the nonlinear marginal effect of variable i.
- $f_{ij}(\cdot,\cdot)$ captures pairwise interaction effects.
- Additive structure makes the model inherently interpretable.



Residual→（x1·x2）选择最能降低RSS的40组交乘项，再进行EBM

H spread = 0.05+0.03-0.01

f(x1·x2):H spread+uncertainty ↑；L spread 不受影响

Design–Variable Importance

- **Mean absolute score:**

$$S(i) = \sum_{(l,t)\in\mathcal{T}_1} w_{l,t} \left| f_i(x_{i,l,t}) \right| .$$

  - The first measure captures the average magnitude of a predictor's contribution.

- **Drop in model fit:**

$$\Delta R_i^2 = R_{-i}^2 - R^2.$$

  - The second measure captures how much predictive fit is lost when the term is removed.

Design–Data and Sample

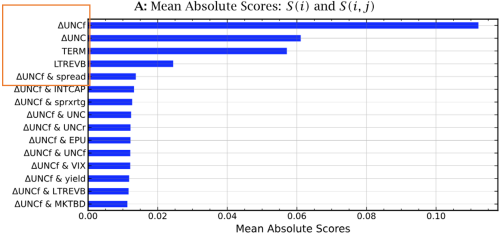
- **Sample:** U.S. corporate bond market.
- **Period:** July 2002 to August 2020.2002-2010 train,2011 valid.
- **Unit:** firm-month observations.
- **Sample size:** 106,265 firm-date observations.
- **Predictors:** 81 variables, including:
  - bond and firm characteristics;
  - bond and equity market factors;
  - macroeconomic and uncertainty variables.

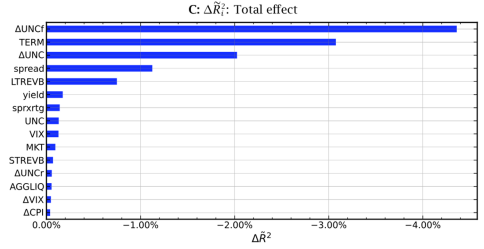
Result–Model Comparison of R2’ s.

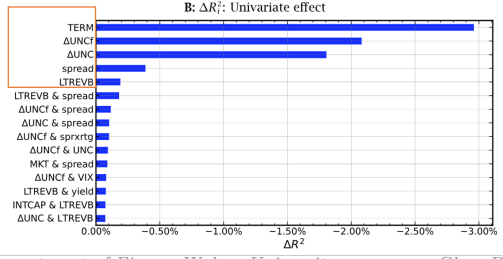
	OLS	LASSO	R-Forest	XGBoost	EBM
R^2_{train}	31.15%	17.51%	46.93%	42.52%	24.79%
R^2_{valid}	−0.27%	15.94%	15.57%	21.80%	18.54%
R^2_{oos}	−2.34%	7.99%	13.34%	11.97%	12.10%

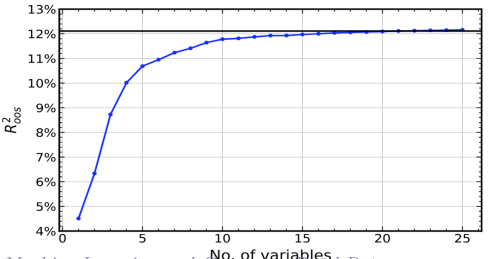
Note: This table shows R^2’s of OLS, LASSO, Random Forest, XGBoost, and EBM estimations in the training (July 2002–June 2010), validation (July 2010–June 2011), and testing (July 2011–August 2020) samples.

Result–Importance of predictors

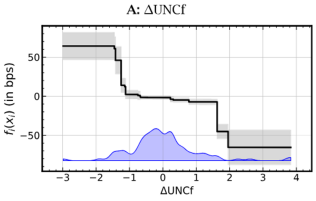


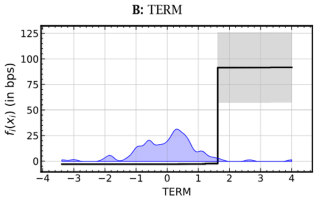


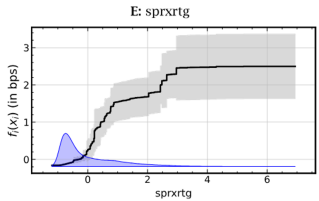


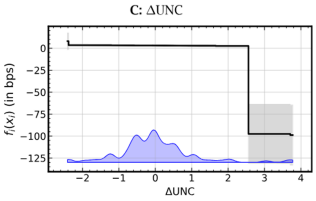


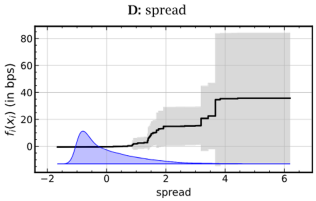
Result–Univariate effects



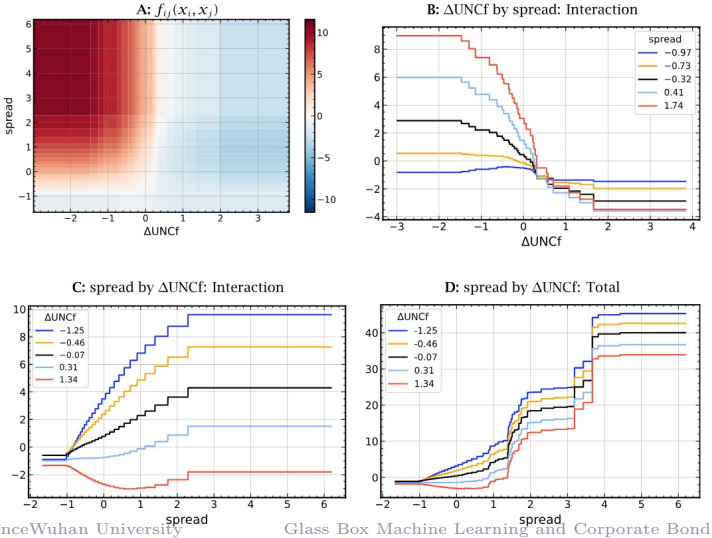








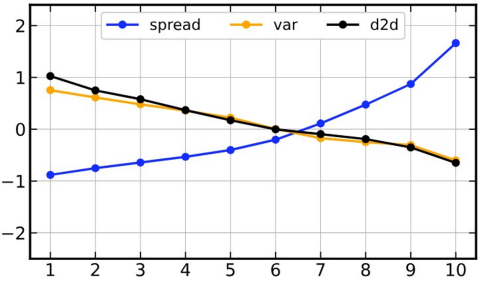
Result–Interaction effects



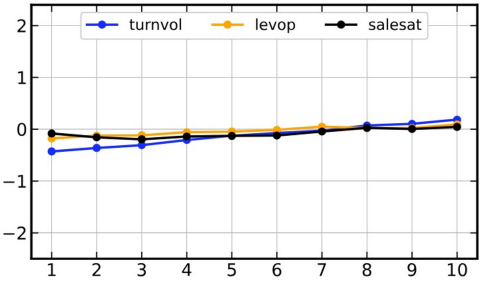
Result–Portfolios based on the EBM model

	1	2	3	4	5	6	7	8	9	10	10-1
					Panel A: EBM VW						
Mean	−0.06	0.04	0.09	0.12	0.16	0.21	0.21	0.28	0.37	0.61	0.67
S.D.	1.28	1.22	1.12	1.25	1.37	1.50	1.71	1.92	2.09	2.60	1.75
SR	−0.05	0.03	0.08	0.10	0.12	0.14	0.12	0.14	0.18	0.23	0.38

A: High scores $S(i)$



B: Low scores $S(i)$



Result–EBM V.S. SHAP

Top 10 variables.

	EBM	Random Forest	XGBoost
1	ΔUNCf	ΔUNCf	ΔUNCf
2	ΔUNC	levmkt	spread
3	TERM	ΔUNC	UNC
4	LTREVB	TERM	ret
5	ΔUNCf, spread	UNC	ΔUNC
6	ΔUNCf, INTCAP	spread	ΔUNCf, spread
7	ΔUNCf, sprxrtg	ΔUNCr	ΔUNCf, UNCf
8	ΔUNCf, UNC	LTREVB	VIX
9	ΔUNCf, UNCr	EPUT	ΔUNCf, UNC
10	ΔUNCf, EPU	VIX	MOMB

SHAP: $\hat{r}_{l,t+1} = \phi_0 + \sum_{i=1}^{N} \phi_{i,l,t} + \sum_{i>j} \phi_{ij,l,t},$

Ex: ΔUNCf:

For $x_{i,l,t} = -3.62$ in May 2020, $f_i(x_{i,l,t}) = 64.3$,

RF and XGB SHAP values in May 2020 range from 45.5 to 78.7, and 114.9 to 133.8

Top 10 effects — Comparison of EBM and SHAP.

	EBM vs. RF	EBM vs. XGB	RF vs. XGB
	Panel A: Correlations		
ΔUNCf	0.9654	0.8976	0.9046
ΔUNC	0.9953	0.9992	0.9954
TERM	0.9812	0.9726	0.9707
LTREVB	0.9708	0.6651	0.7488
ΔUNCf & spread	0.3566	0.4147	0.5605
ΔUNCf & INTCAP	0.4930	0.2124	0.4994
ΔUNCf & sprxrtg	0.3573	–	–
ΔUNCf & UNC	0.4909	0.5184	0.7100
ΔUNCf & UNCr	0.3695	−0.0334	−0.2122
ΔUNCf & EPU	0.0830	−0.3423	0.1679

Decompositions of effects.

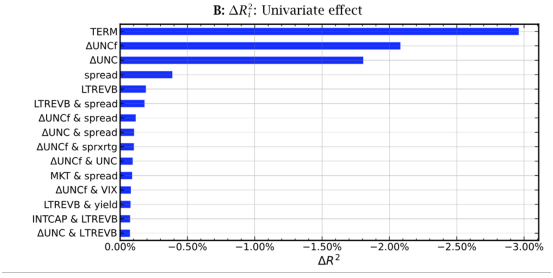
Variables	EBM	RF	XGB
Univariate	0.11	0.34	0.42
Interactions	0.89	0.66	0.58
Firm	0.17	0.17	0.18
Mkt	0.36	0.53	0.54
Firm/mkt	0.47	0.30	0.28
Univariate-firm	0.01	0.09	0.13
Univariate-mkt	0.10	0.25	0.29
Interaction-firm	0.16	0.08	0.05
Interaction-mkt	0.26	0.29	0.25
Interaction-firm/mkt	0.47	0.30	0.28

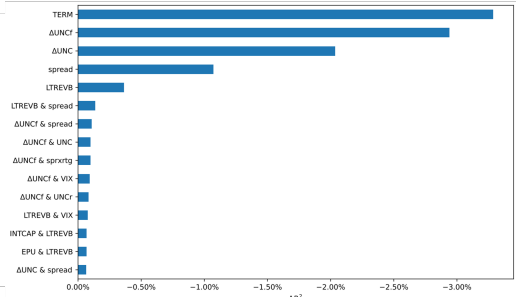
Replication1

复现：

	OLS	LASSO	R-Forest	XGBoost	EBM
R^2_{train}	31.15%	17.51%	46.93%	42.52%	24.79%
R^2_{valid}	−0.27%	15.94%	15.57%	21.80%	18.54%
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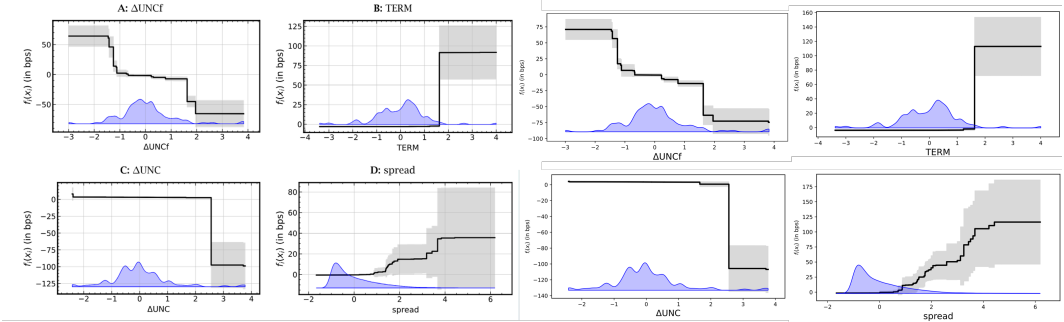
	OLS	LASSO	EBM
R2train	31.09%	17.49%	25.96%
R2valid	-2.43%	15.93%	17.66%
R200S	-2.50%	7.98%	13.06%



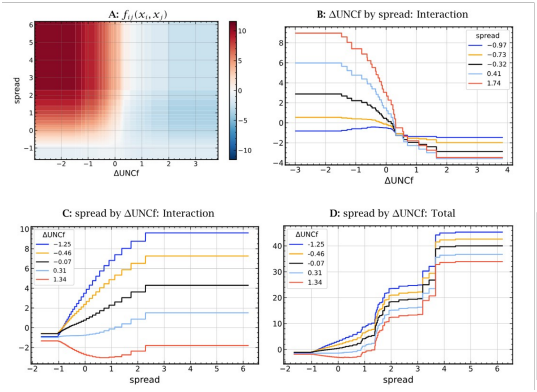


Replication2

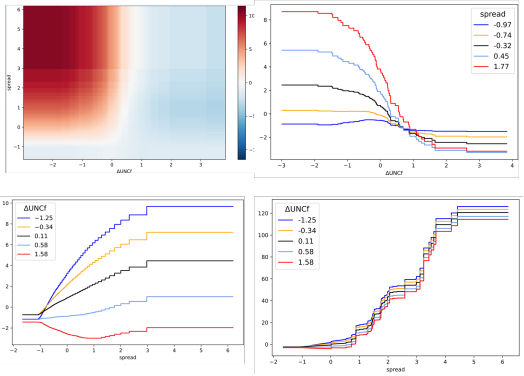
复现：



Replication3



复现：



Thanks!

Question & idea?

Idea

- 中国债券市场应用，探寻宏观变量与特征的影响解释
- 其他预测场景：企业风险预警

Appendix 1: EBM Algorithm I

Table B.1: Hyperparameters

Model	LASSO	Random Forest	XGBoost	EBM
Parameters	$\alpha \in (10^{-3}, 10^{-1})$	$\#Trees = 100$ *Min Samples Leaf* $\in \{1, 2, 4\}$ *Min Samples Split* $\in \{2, 4, 8\}$ *Max Features* $\in \{0.1, 0.25, 0.5, 0.75, 1\}$ *Max Depth* $\in \{2, 4, 8\}$	$\#Trees \in \{10, 50, 100\}$ *Max Depth* $\in \{5, 6, 7, 8, 9\}$ *Learning Rate* $\in \{0.01, 0.1, 0.3, 0.5\}$ $Subsample \in \{0.5, 0.75, 1\}$ $Lambda \in \{1, 2, 4, 8\}$	*Early Stopping Rounds* $\in \{10, 20\}$ *Early Stopping Tolerance* $\in \{0.01, 0.005\}$ *Smoothing Rounds* $= 25$ *Interaction Smoothing Rounds* $\in \{250, 500\}$ *Greedy Ratio* $\in \{1.5, 5\}$ *Learning Rate* $= 0.00025$ *Max Leaves* $= 2$ *Max Rounds* $= 500$ *Min Hessian* $= 0.00001$ *Min Samples Leaves* $\in \{2, 20\}$ $Interactions \in \{30, 40\}$ *Inner Bags* $\in \{0, 5\}$ *Outer Bags* $= 10$ *Validation Size* $\in \{0.5, 0.75\}$